MILITARY SPECIFICATION

GENERATOR, GAS

Bureau of Weapons Drawing 2301891

(For Parachute Release Mk 32 Mod 0)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of Gas Generator referred to herein as the "Generator". See 6.1 for description of intended use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.- Unless otherwise stated the following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5272	Environmental testing, Aeronautical and Associated Equipment, General Specification for

FSC -1350

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS 16625	Ring, Retaining, Internal, Basic (Tapered Section Type)
MS 33586	Metals, Definitions of Dissimilar

DRAWINGS

Bureau of Naval Weapons

2068881	Gasket
2159752	Spacer
2301891	Generator, Gas Specification Control Drawing
2452767	Retainer
2551270	closure

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications .- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

PUBLICATIONS

Interstate Commerce Commission

49 CFR, Part 71 to 78	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

2.3 Conflicting requirements.-

Unless otherwise specified when conflicting statements exist within the documentation for the Generator, the following shall be the order of precedence:

A. Contract or Purchase Order

B. Drawings originated specifically for the Generator

C. This Specification

D. Federal Documents

E. Military documents

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified in the contract or order, prior to the start of production of Generators, a preproduction sample shall:

(a) Be manufactured using the procedures proposed for production
(b) Consist of 50 Generators
(c) Be tested as specified in section 4 herein and is for the purpose of determining that, prior to starting production, the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract.

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3.2 Basic requirements. -

3.2.1 Data. - No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2 unless specified in the contract or order (see paragragh 6.2).

3.2.2 Selection of materials.-

3.2.2.1 Fungus proof material.- Materials that are nutrients for fungi shall not be used unless necessary. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.2.2.2 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in Standards 33586.

3.2.3 Design changes.- Unless otherwise specified by the procuring activity, any change in design (which affects performance or reliability of the Generator as determined by the Procuring Activity) made after completion of preproduction tests, shall require that a new preproduction sample (see 4.3.1) manufactured in accordance with the design change, be submitted for testing.

3.2.4 Gages.- The supplier shall provide whatever gages are necessary and adequate to assure that the material and parts of the Generator will meet dimensions and requirements shown on the applicable documents listed on BuWeps Drawing 2303.891.. When requested by the supplier, the procuring activity will furnish existing pertinent drawings of Naval Final inspection gages for use only in the designing of the supplier's gages. This action does nut relieve the supplier of the responsibility for designing and manufacturing the gages needed for the satisfactory fulfillment of the contract requirements.

3.2.5 Inspection and test equipment and facilities.- The supplier shall furnish and maintain all necessary inspection and test equipment and facilities, and provide personnel for performing all acceptance tests specified herein. The test equipment shall be adequate to permit performance of the required acceptance tests.

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3.2.6 Safety. - The supplier or manufacturer, as may apply, shall provide adequate safety precautions for protection of personnel during the handling of the Generator. See General Safety precautions, 6.4.

3.3 Performance and physical characteristics .-

3.3.1 Initiation. - The Generator shall be mechanically initiated by the extraction of the firing sear.

3.3.1.1 Extraction. -

3.3.1.1.1 Force and distance.- The Force, applied at any angle within the force extraction angle (see BuWeps dwg 2301891) required to extract the firing sear shall be within the range of 5 to 20 pounds (lbs) acting for a distance of 0.50 ± .25 inches.

3.3.1.1.2 Velocity.- The Generator shall allow extraction of the firing sear at any velocity up to and including 200 feet per second (fps).

3.3.2 Pressure from firing.- The gas pressure from the Generator when fired into a free volume (excluding voids within the Generator) of 0.50 ± .01 cubic centimeters (cc) shall have a time-pressure curve within the shaded area of figure 1.

3.3.3 External pressure.- The capped end of the Generator (end opposite the firing sear) shall withstand 3700 pounds per square inch gage (psig) gas pressure minimum for 10 seconds minimum without leakage.

3.3.4 Safety pin.- The Generator shall incorporate a removable safety pin that prevents movement of the firing mechanism when safety pin is in place.

3.3.5 Configuration.- The envelope dimensions, part identification, mounting, and other external dimensions shall be in accordance with 3.5.5.1 and the applicable specification control drawings (BuWeps Drawing 2301891).

3.3.5.1 Firing Sear.- The firing sear shall be capable of passing through a .145 dia hole during extraction and separation, at any angle within the sear extraction cone (see BuWeps Dwg 2301891 and figure 3) while conforming to 3.3.1.

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3.4 Environmental requirements. - Unless otherwise specified, the Generator shall be capable of conforming to the performance requirements and physical characteristics of this specification after being exposed to the environments and all conditions specified in this section.

3.4.1 Vibration (see 6.3). -

3.4.1.1 Ready condition.- The Generator shall be capable of withstanding the vibration shown in Table I in the Ready condition (see 6.3).

Table 1

Double Amplitude (in. minimum	Acceleration (g) minimum	Frequency Range (cps)
0.10	- -	16-24
- -	3.0	24-40
0.036	- -	40-74
- -	10.0	74-500

3.4.1.2 Packaged.- The Generator packaged in accordance with 5.1 shall be capable of withstanding the vibration of Table II.

Table II

Double Amplitude (in.) minimum	Acceleration (g) minimum	Frequency Range (cps)
0.75	- -	5-6
- -	1.3	6-60

3.4.2 Shock (see 6.3).- The Generator shall be capable of withstanding shock specified in Table III.

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Table III

Shock Schedule

Condition (see 6.3)	Peak Magnitude (g)	Minimum Duration (ms)	Acceleration Direction
Nonoperating	60	8.0	Any
Ready	45	20.0	Forward

3.4.3 Acceleration .- The Generator shall be capable of withstanding an acceleration of 35 g's in the aft direction and 25 g's in the forward direction with the Generator in the ready condition (see 6.3).

3.4.4 Temperature.-

3.4.4.1 Operating.- The Generator shall meet the requirements of 3.3 at any temperature from 0 to 140 degrees Fahrenheit (°F).

3.4.4.2 Not operating.- The Generator shall meet the performance requirements and product characteristics after storage at any temperature from -45°F to +140°F upon stabilization at any temperature specified in 3.4.4.1.

3.4.5 Humidity.- The Generator shall withstand exposure to 95 percent relative humidity at 110°F for eight hours without degradation of finish or functional capabilities.

3.4.6 Altitude.- The Generator shall meet the performance requirements of 3.3 subsequent to exposure to atmospheric pressure at any altitude from 0 to 20,000 feet.

3.4.7 Attitude.- The Generator shall meet the performance requirements of 3.3 in any attitude.

3.5 Life.-

3.5.1 Storage life, ready nonoperating.- The Generator, mounted in service installation, shall have a storage life of one year.

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3.6 Identification.. The Generator shall be permanently identified in accordance with the applicable documents listed in BuWeps Drawing 2301891.

3.7 Workmanship.- The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All. parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Test conditions.- Unless otherwise specified, the Generator shall be subjected to tests under the following conditions:

(a) Temperature 60 to 95 °F (room ambient)

(b) Humidity (relative) Room ambient, up to 95 percent

(c) Altitude Normal ground atmospheric pressure

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4.3 Classification of inspection.- Inspection of the Generator shall be as follows:

(a) Preproduction inspection (4.4)

(1) Preproduction sample (see 4.4.1)

(2) Preproduction tests (see 4.4.2)

(b) Periodic production sampling (see 4.5)

(1) Periodic production sampling (see 4.5.1)

(2) Periodic production tests (see 4.5.2)

(c) Acceptance inspection (4.6)

(1) Acceptance sampling (see 4.6.1)

(2) Acceptance tests and examination (see 4.6.2)

4.4 Preproduction inspection

4.4.1 preproduction sample.- Prior to the start of production, a preproduction sample of 50 Generators manufactured in accordance with 3.1, shall be subjected to the pre-production tests detailed in 4.4.2 at a facility denigrated by the procuring activity. Acceptance of the preproduction sample shall be based on no detects in the sample. Further production of the Generator by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.4.2 Preproduction tests.- The Generators within the preproduction sample (identified as Generators numbers 1 through 50) snall be subjected to the tests as specified in Table IV. Certified inspection data covering the results of preproduction tests shall be forwarded to the Bureau of Naval Weapons. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests

(b) Run or test plan used

(c) Serial numbers or identification of the Generator

(d) Test results of preproduction tests

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TABLE IV

Preproduction Test Schedule

Test Description		Generator Numbers (see 4.4.2)																									
		1 2	3 4	5 6	7 8	9 10	11 12	13 14	15 16	17 18	19 20	21 22	23 24	25 26	27 28	29 30	31 32	33 34	35 36	37 38	39 40	41 42	43 44	45 46	47 48	49 50
Examination Inspection (4.6.2.1)		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Leakage (4.4.2.2)		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Vibration (4.4.2.1.2.1.)		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x									
Vibration (4.4.2.1.2.2)												x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Shock (4.4.2.1.3.1)		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x									
Shock (4.4.2.1.3.2)												x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Acceleration (4.4.2.1.4)		x	x				x			x	x	x	x	x	x	x	x	x	x			x	x			
Temperature Cycling (4.4.2.1.5)		x	x			x	x			x	x	x	x	x	x	x	x									
Humidity (4.4.2.1.6)						x						x	x	x	x	x	x					x				
Altitude (4.4.2.1.7)				x								x	x	x	x	x	x					x				
Function (4.4.2.3)	0°F	x		x	x		x					x	x												x	
	70°F									x	x			x	x			x	x	x	x	x	x	x		
	140°F		x			x		x	x							x	x									x

4.4.2.1 Environmental tests .- The Generators shall be subjected to the environmental tests (see Table IV)- Unless otherwise specified, inability of the Generator to pass the leakage and function tests of 4.4.2.2 and 4.4.2.3 subsequent to the environmental tests, constitutes a failure.

4.4.2.1.2 Vibration.-

4.4.2.1.2.1 Installed.- Mount the Generator, with the safety pin removed, as shown in figure 2. The Generator shall be conditioned at room temperature for a minimum period of 1 hour prior to conducting the resonance surveys. vibrate the unit at room temperature for a total of 9 hours, in accordance with Procedure XII, Specification MIL-E-5272C, and Table I of 3.4.1.1.

4.4.2.1.2.2 Packaged.- Mount the packaged Generator on the vibration test equipment. The packaged Generator shall be vibrated at *room* ambient temperature. Cycle the applied vibration from 5 to 60 *cps* in accordance with Table II of 3.4.1.2 at the rate of 1 cycle (from 5 to 60 cps and return) every 5 minutes for 15 minutes in each of three mutually perpendicular axis.

4.4.2.1.3 Shock.- The Generator shall be subjected to shock at room temperature (4.4.2.1.3.1 and 4.4.2.1.3.2) in accordance with the shock test procedure IV of Specification MIL-E-5272C except shock magnitude and duration shall be measured using an accelerometer mounted on the input table of the fixture to indicate applied shock, and the test equipment is not restricted to that specified in Specification MIL-E-5272C.

4.4.2.1.3.1 Safety pin installed.- Mount the Generator as shown in figure 2 except the safety pin shall be installed. Secure to the shock test fixture so that a minimum of 90 percent of the table input of the fixture will be applied to the Generator. Subject the Generator to a total of 9 shocks of 60 g each for 8 +2 -O milliseconds. Two shocks, one in each direction, shall be applied along the mutually perpendicular principle axis and three shocks shall be applied in any direction.

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4.4. 2.1.3.2 Installed. - Mount the Generator with safety pin removed, as shown in figure 2 and secure to the shock test fixtures so that a minimum of 90 percent of the table input of the fixture will be applied to the Generator. Subject the Generator to 45 g's, peak magnitude for a duration of 20.0 +5.0 -0 milliseconds in the forward direction (see figure 2).

4.4.2.1.4 Acceleration.- The Generator, with safety pin removed, shall be mounted as shown in figure 2 and subjected to an acceleration of 35 g's in the forward direction for 3 minutes, and 25 g's in the aft direction for 3 minutes at room temperature.

4.4.2.1.5 Temperature cycling.- The Generator, preserved and packaged in accordance with section 5, shall be placed in a temperature controlled chamber and subjected to the following chamber temperature cycle:

(a) Cool to -45 -0 +5 degrees l? within 30 minutes
(b) Maintain at -45 -0 +5 degrees F for 4 hours
(c) Heat to 140 +0 -5 degrees F within 30 minutes
(d) Maintain at 140 +0 -5 degrees F for 4 hours
(e) Return to room temperature within 15 minutes

Repeat this schedule for three cycles.

4.4.2.1.6 Humidity.- Mount the unpackaged Generator with safety pin removed, in the humidity chamber and subject it to an atmosphere of 95 +5 -0 percent relative humidity at 110 +5 -0 F for 8 +1 -0 hours.

4.4.2.1.7 Altitude.- Place the unpackaged Generator with safety in removed, in a vacuum chamber and reduce the pressure to 13.76 inches of mercury absolute.

4.4.2.2 Leakage.- Install the Generator to be tested in the test fixture (see figure 3). Subject the cartridge end (opposite the firing sear) to 3700 +50 -0 psig air pressure at ambient temperature for 10 +1.0 -0 seconds. A decrease of pressure due to leakage through the Generator during the test constitutes a failure.

4.4.2.3 Function tests.- The Generator shall be installed on the test fixture in the test chamber (see figure 3) and be initiated by applying a force in accordance with 3.3.1 at random extraction cone angles in accordance with 4.4.2.3,1. One half of the preproduction sample shall be fired with their longitudinal axis (see figure 1) horizontal; thirteen shall be fired with the firing sear end up and the remaining Generators shall be fired with the firing sear end down.

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4.4.2.3 (Cont.)

The Generators to be fired at 140°F (see Table IV) shall be conditioned to 140 +0 -5°F for 1 hour. The Generators to be fired at 0 °F shall be conditioned to 0 +5 -0°F for 1 hour, and the remaider shall be fired at room ambient temperature. All Generators shall be fired within 4 minutes after one hour temperature conditioning period. Inability of the Generator to meet the requirements of 3.3 2 constitutes a failure.

4.4.2.3.1 Extraction Cone angles . - The Generator samples shall be divided to allow sear extraction to be accomplished at azimuth angle increments of 30°, at angles from the Generator longitudinal axis of 0 , 30 , 45 and 60 . The azimuth plane is perpendicular to the longitudinal axis with the safety pin axis used as O reference.

4.4.2.3.2 Extraction velocities. - The sear extraction velocities required by 3.3.1.1.2 may be achieved by using a triggered snatch hook mounted on a centrifuge and the Generator held near the periphery.

4.4.3 Preproduction test equipment.- The following items of test equipment, or equivalent items, will be required to perform the tests specified in 4.4.2.

(a) Vibration test equipment capable of applying vibration as specified in 4.4.2.1.2

(b) Shock test equipment capable of applying shock as specified in 4.4.2.1.3

(c) Acceleration test equipment capable of acceleration as specified in 4.4.2.1.4

(d) Chamber, temperature-humiiity capable of providing environments as specified in 4.4.2.1.5 and 4.4.2.1.6

(e) Test fixture - see figure 3

(f) Pressure air supply 3700 psig minimum

(g) Vacuum chamber capable of 13 inches of Mercury absolute.

(h) Test fixture (see figure 2).

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4.5 Periodic production inspection .-

4.5.1 Periodic production sampling.- Periodic production tests shall be performed at a facility designated by the procuring activity. Samples shall consist of units that have satisfactorily passed the acceptance examinations of 4.6, and shall be selected as follows:

(a) Step 1.- Upon initiation of production, randomly select 1 unit from each group of 10 units. Upon satisfactory compliance of samples selected from three consecutive groups, proceed to step 2.

(b) Step 2.- Randomly select 1 unit from each group Of 25 units. Upon satisfactory compliance of samples selected from three consecutive groups, proceed to step 3.

(c) Step 3.- Randomly select 1 unit from each group of 50 units. Continue with step 3 for the balance of the contract or until a defective unit is found.

4.5.1.1 Defective units.- When a defective unit is found in any group, the supplier shall:

(a) Determine the cause of the defect

(b) Determine if the defect is an isolated case

(c) Take the necessary action to eliminate the defect from units on hand, and, if so directed by the procuring activity, from those units previously shipped.

(d) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

(e) Return to step 1 (4.5.1(a)) and proceed as indicated for inspection of units subsequently produced.

4.5.2 Periodic production tests.- The periodic sample, shall be subjected to the following tests: 4.5.2.1, 4.5.2.2, 4.5.2.3, and 4.5.2.4. Certified inspection data concerning the results of the periodic production tests shall be forwarded to the Bureau of Naval Weapons. The inspection data shall consist of the following:

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(a) Date of tests and names of persons conducting test

(b) Run or test plan used

(c) Serial number or identification number of the Generators

(d) Test results of the periodic production tests.

4.5.2.1 Vibration 4.4.2.1.2.1

4.5.2.2 Shock 4.4.2.1.3

4.5.2.3 Leakage 4.4.2.2

4. 5.2.4 Function tests. - The Generator shall be subjected to the function test of 4.4.2.3 at room ambient temperature with the longitudinal axis in the horizontal plane

4.5.3 Periodic production test equipment.- The following items of test equipment or equivalent will be required to perform the periodic production tests:

(a) See 4.4.3(a)

(b) See 4.4.3(b)

(c) See 4.4.3(e)

(d) See 4.&.3(f)

4.6 Acceptance inspection.- Acceptance inspection consists of two parts, as follows:

(a) Examination

(b) Lot acceptance tests

4.6.1 Acceptance sampling.-

4.6.1.1 Examination. - The acceptance examinations shall be conducted on each Generator to be delivered.

4.6.1.2 Lot acceptance. - Unless otherwise specified in the contract or order, a random sample of Generators shall be selected from each inspection lot for the testing required for lot acceptance (see 4.6.2.2). The size of this sample shall conform to the requirements of Standard MIL-ST9-105 for an Acceptance Quality Level (AQL) of 0.25%, except the sample size shall never be less than 37 Generators.

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4.6.2 Acceptance tests and examinations. -

4.6.2.1 Examinations. - The Generator shall be examined for conformance to the requirements herein and on BuWeps drawing 2301891. The non-conformance to the dimensions or requirements on BuWeps drawing 2301891 or the presence of any of the following constitutes a failure:

(a) Evidence of reclaimed, used, or *salvage* parts

(b) Safety pin not installed

(c) Dents, scrapes, nicks, or other damage as determined by procuring activity

(d) Evidence of parts missing

(e) Evidence of improper assembly

(f) Evidence of loose cap

(g) Preservation-and packaging not as specified in 5.1.

4.6.2.2 Tests.- The samples selected in accordance with 4.6.1.2 shall be subjected to the tests of 4.6.2.2.1 and 4.6.2.2.2.

4.6.2.2.1 Leakage. - 4.4.2.2

4.6.2.2.2 Function tests.- The Generator shall be subjected to the function tests of 4.4.2.3 at room ambient temperature with the longitudinal axis in the horizontal plane.

4.6.2.3 Acceptance test equipment.- The following items of test equipment or equivalent will be required to perform the acceptance tests.

(a) See 4.4.3(e)

(b) See 4.4.3(f)

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5. PREPARATION FOR DELIVERY

5.1 Preservation and -packaging .- Preservation and packaging of the Generator shall be level A. The unit package shall be one Generator.

5.1.1 Level A.- The Generator shall be preserved and packaged in accordance with Code of Federal Regulations 49 CFR 73.106.

5.1.2 Level B.- Not applicable.

5.1.3 Level C.- Not applicable.

5.2 Packing.- Packing of the Generator shall be level A.

5.2.1 Level A.- The Generator, preserved and packaged in accordance with 5.1.1, shall be packed in a container in accordance with the Code of Federal Regulations 49 CFR 73.106.

5.2.2 Level B.- Not applicable.

5.2.3 Level C.- Not applicable.

5.3 Marking.- Unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129 and the Code of Federal Regulations 49 CFR 73.106.

6. NOTES

6.1 Description and intended use.-

6.1.1 Description. - The Gas Generator contains pyrotechnic material initiated by a delay fuze which in turn has been initiated by a “match-head” fired by extraction of a firing sear.

6.1.2 Intended use - The Generator supplies gas pressure to actuate a piston for an unlocking device in the Mk 32 Mod O Parachute Release.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

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(b) Serialization

(c) Packing Quantity

(d) Assigned activity for preproduction and periodic tests

(e) All additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed periodic production inspections

(f) Data required other than that specified in 3.3.1

(g) Any exceptions to this specification.

6.3 Definitions. -

6.3.1 Generator Condition.-

6.3.1.1 Nonoperating. - The condition of the Generator (with safety pin installed) when stored, transported, or handled.

6.3.1.2 Ready, nonoperating and ready.- The condition of the Generator when assembled with the end item (safety pin removed), transported, and handled.

6.3.1.3 Operating.- Operating is the condition of the Generator when operating.

6.3.2 Vibration.- Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak to peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (cps).

6.3.3 Shock.- Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), duration in units of time, milliseconds (ms), and a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent at maximum slope, intersect the "zero" g line.

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6.3.4 Acceleration (steady -state).- Acceleration (steady-state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g's).

6.3.5 Humidity.- The relative humidity of the environment to which the Generator is exposed, expressed as a percent.

6.4 General safety precautions.- The loading, assembly and handling of the Generator and the subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification will not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are certain minimum provisions which a contractor or manufacturer (which explosively loads the item covered) should observe in order to fulfill his responsibility for safety. At Naval Ordnance Systems Command. and other Government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail technical requirements of the specifications. Personnel involved in these operations shall be indoctrinated in these safety practices. The procedures of BuWeps Publication OP 5, Ammunition Ashore, Handling, Stowing, and Shipping, shall be followed where applicable.

6.4.1 All loading operations should be conducted in a neat and orderly manner.

6.4.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

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6.4.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning, should avoid, insofar as practicable, using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.4.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.4.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with intraplant distances when outside of loading rooms. For Navy managed explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives shall apply.

6.4.6 proper care shall be exercised at all times to protect personnel from accidents, fires or explosions, or to limit damage to equipment and loading areas. In this connection, the precautionary measures in the following paragraphs should be observed.

6.4.6.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.4.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operations.

6.4.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of *or* put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.4,6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and mixtures; and where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and worker's conductive shoes. Worker's clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation, should be avoided. Additional ground

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devices such as grounded bracelets for workers should be employed where operations are conducted with items whicn are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare ncn-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heat apparatus, etc. Where necessary for safety, humidity of work room should be appropriately increased, as required to lessen electrostatic effects but without inducing excessive moisture absorption by any of the components of the item being loaded.

6.4 6.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., insulation, grounding separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric-type primers, detonators, or squibs provided with the wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips or other devices should be employed during manufacturing operations involving electric primers, detonators, or squibs.

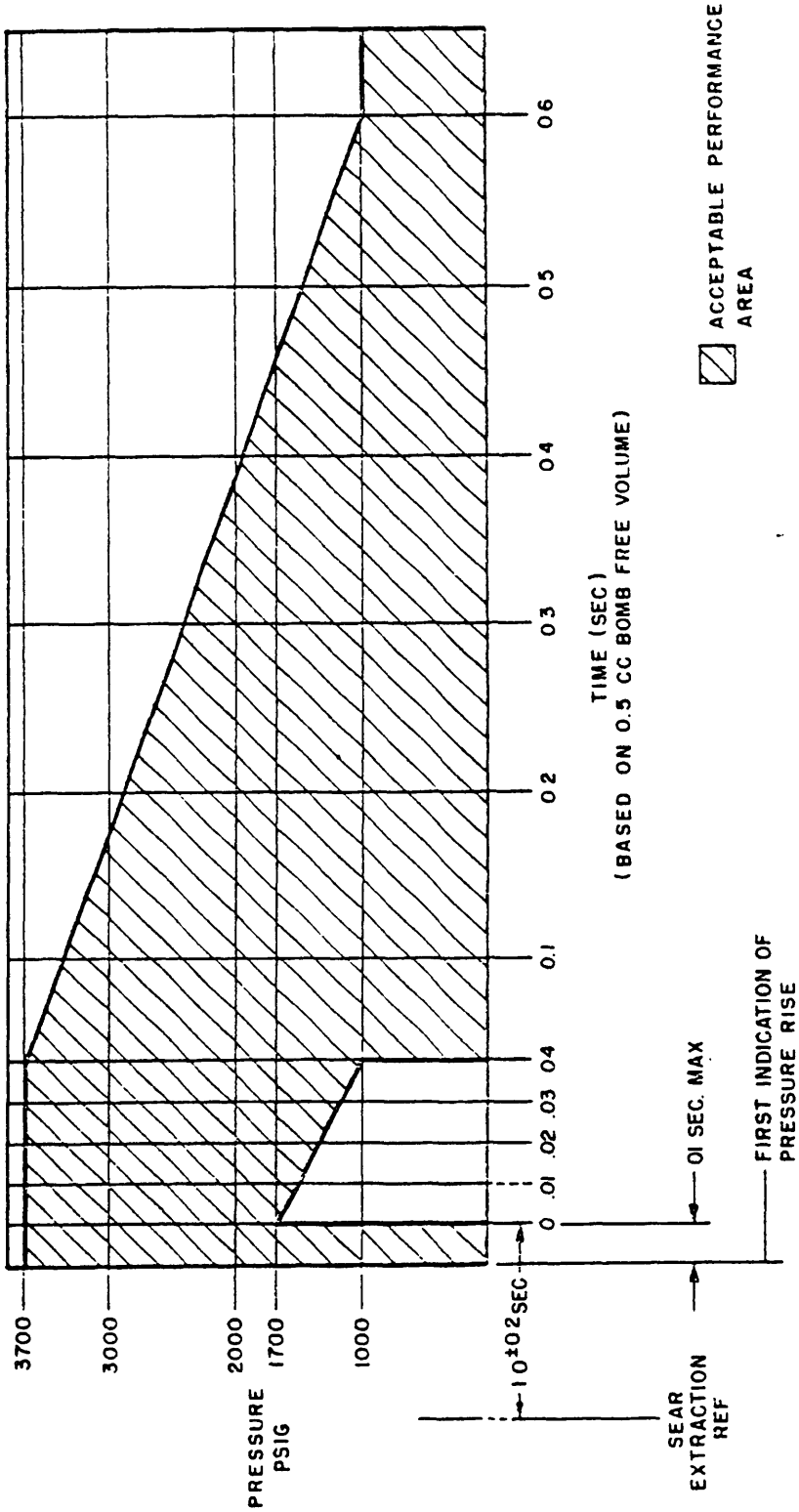
6.4.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

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6.4.6.7 Allow no fires or exposed electrical or other sparking equipment and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper “Match” and “No Smoking” rules where necessary.

6.4.6.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

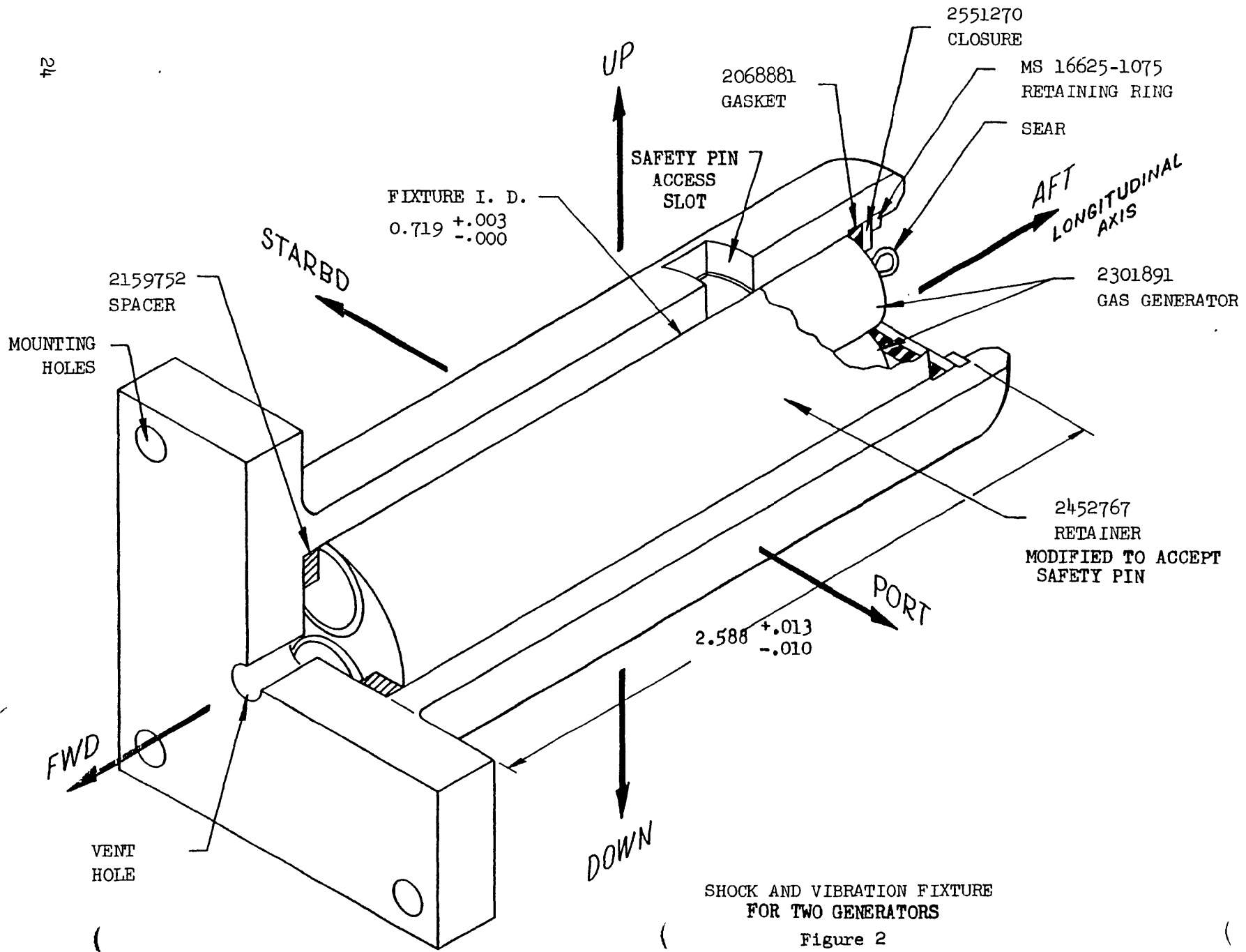
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FIRING PRESSURE - TIME CURVE

Figure 1

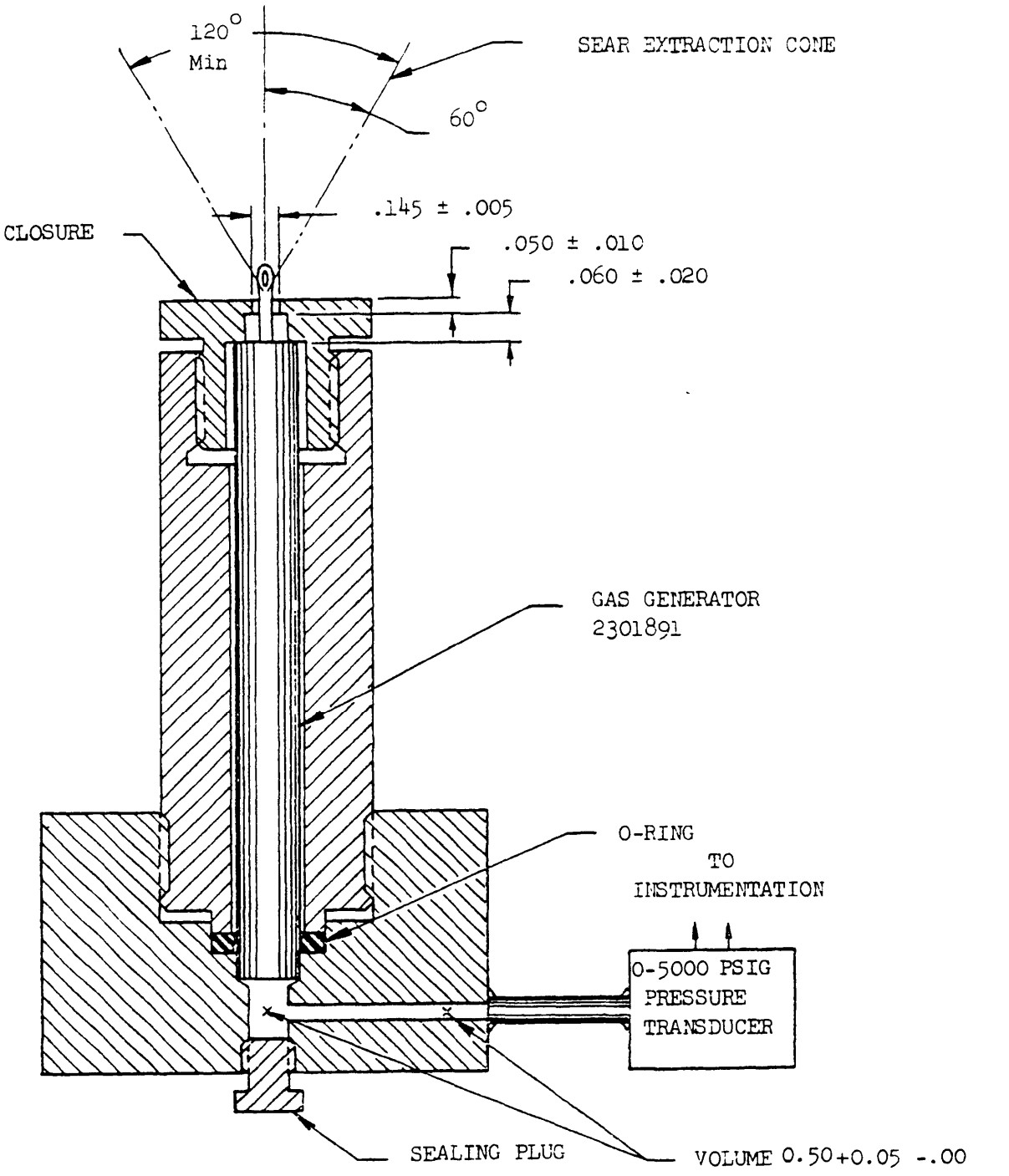
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SHOCK AND VIBRATION FIXTURE
FOR TWO GENERATORS

Figure 2

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LEAKAGE AND FUNCTION TEST FIXTURE

Figure 3

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-G-82455(OS) GENERATOR GAS (FOR PARACHUTE RELEASE MK #32 Mod 0)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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